Grey Wolf Optimizer and Deep Neural Network based Feature Selection and Classification in Medical Image Analysis

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Abstract— Accurate and efficient medical image analysis is required for modern healthcare applications, facilitating timely disease diagnosis and treatment. Considering the importance of optimization algorithms and deep learning methodologies, this work presents a novel approach, Grey Wolf Optimizer-Assisted Deep Neural Networks (GWO-DNN), for feature selection and classification in medical image analysis. The proposed framework utilizes the power of the Grey Wolf Optimizer to intelligently select the most relevant features from complex medical images. The selected features are then integrated into Deep Neural Networks, to create accurate models for medical image classification tasks. This integrated approach is designed to improve diagnostic accuracy and enhance patient care. The experimentation and evaluation, reveal that the proposed GWO-DNN based model for image analysis is more efficient than the existing methods.

Keywords— Grey Wolf Optimization (GWO), Deep Neural Networks (DNN), Medical Image Analysis, Feature Selection, Disease Classification

I. INTRODUCTION

Medical image analysis plays a critical role in healthcare, enabling early detection, diagnosis, and treatment of wide range of diseases. Advanced machine learning techniques are required to extract meaningful information from images and improve clinical decision-making [2]. In this perspective, fusion of optimization algorithms and deep learning methodologies has emerged as a promising approach to improve the accuracy and efficiency of medical image analysis. introduces a framework that integrates Grey Wolf Optimizer (GWO) with Deep Neural Networks (DNNs) for feature selection and classification in medical image analysis. GWO, considers behavior of grey wolves, and offers a powerful optimization mechanism that guides the selection of relevant features from complex medical images. By integrating GWO with DNNs, critical challenges in medical image analysis can be addressed by the collective intelligence of these algorithms.

The objectives of the proposed GWO-DNN are twofold: first, to develop an efficient feature selection mechanism using GWO that identifies most informative regions and features within medical images, reducing dimensionality and preserving diagnostic information. Second, to integrate these selected features into a DNN architecture for accurate classification of medical images, facilitating disease identification. Thus, GWO and DNN offers a significant approach to feature selection and classification, with applications such as radiology, pathology, and more Additionally, the proposed work could make healthcare more cost-effective by reducing the need for time-consuming and expensive manual analysis.

II. BACKGROUND WORK

The background work highlights the significance of addressing feature selection challenges in medical image analysis, specifies the potential of GWO as an optimization tool, and introduces the need for combining GWO [16-17] with DNNs in the proposed approach. Medical image analysis has undergone significant advancements in recent years, due to both machine learning techniques and the availability of large-scale medical imaging datasets. However, medical image analysis remains a complex task due to the inherent challenges associated with the diversity and dimensionality of medical image data [1-5]. Deep Learning architectures, specifically Convolutional Neural Networks (CNNs), have emerged as a dominant technology in medical image analysis. CNNs demonstrated remarkable capabilities in automated feature extraction and pattern recognition from medical images, to handle tasks such as tumor detection, organ segmentation, and disease classification [18-24]. Also, deep learning models often require a large number of features to operate effectively, which can lead to computational in- efficiencies and reduced interpretability [25-30]. Feature selection is an important pre- processing step in medical image analysis to overcome these challenges. It involves identifying informative subset of features from a high-dimensional dataset while maintaining classification accuracy. Feature selection

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methods, such as filter, wrapper, and embedded techniques, are not well-suited to handling the complex and nonlinear relationships in medical image data. Optimization algorithms have shown impact in enhancing feature selection and classification tasks [6-15]. The Grey Wolf Optimizer (GWO), considering cooperative hunting behavior of grey wolves, has gained attention for its ability to navigate complex search spaces effectively. By utilizing the collective intelligence of a wolf pack, GWO can potentially improve feature selection by identifying most relevant features within medical images. Integrating GWO with Deep Neural Networks (DNNs) presents a significant approach to handle challenges of medical image analysis, offering the potential for more accurate and efficient disease diagnosis.

III. Methodology

This section, briefs about the steps of the hybrid GWO-DNN Model to perform medical image analysis. The steps for the proposed model for performing image analysis have been specified in Figure 1. The initial step is to gather a dataset consisting of medical image samples to perform analysis. The proposed work utilizes the ChestX-ray14 dataset from Kaggle for processing.

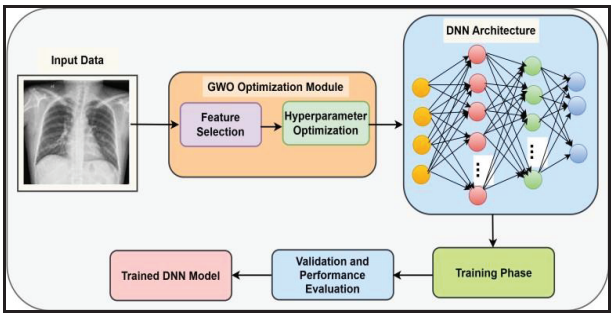


Fig. 1. Medical Image Analysis Framework Using GWO-DNN Architecture

The proposed approach, Grey Wolf Optimizer-Assisted Deep Neural Networks (GWO-DNN), is a hybrid optimization and machine learning technique applied for feature selection and classification in medical image analysis. This approach combines the Grey Wolf Optimizer (GWO), a nature-inspired optimization algorithm, with Deep Neural Network (DNN) for better accuracy and efficiency in medical image classification tasks.The next step is to preprocess the images, using resizing, noise reduction, and intensity normalization, to prepare them for feature extraction. Deep neural networks (DNNs) extract features from preprocessed medical images. Convolutional Neural Networks (CNNs) are used for this purpose. The DNN layers learn hierarchical representations of image features.

The next step is to process Grey Wolf Optimization (GWO) algorithm for optimal feature selection. Initialize grey wolves population. Every wolf specifies a potential solution in the feature selection process. Randomly assign binary values (1 or 0) to each feature to indicate inclusion/exclusion in the feature set. Then apply the GWO algorithm to optimize the feature selection process. Wolves represent different feature sub-sets, and GWO guides the selection of the most relevant features by updating the binary values. The fitness function is used to evaluate the quality of feature subsets considering their impact on classification accuracy. The steps of the proposed GWO-DNN algorithm is specified below.

Algorithm: Grey Wolf Optimizer-Assisted Deep Neural Networks (GWO-DNN)

Inputs:

- Medical image dataset X with corresponding labels Y.
- DNN architecture and hyperparameters.
- GWO parameters (population size, maximum iterations, convergence criteria, etc.)

Outputs: Trained DNN model with optimized hyperparameters and feature selection

Initialization:

1. Initialize a population of grey wolves P with positions randomly.
2. Initialize DNN hyperparameters (e.g., learning rate, batch size, and more).
3. Initialize the maximum number of iterations Tmax.
4. Initialize the current iteration count=0, t=0. Grey Wolf Optimizer (GWO) Phase:
5. while t< Tmax :
   Calculate fitness for each wolf in the population based on feature selection quality:
   - Extract features from medical images using the current wolf positions.
   - Train a DNN model using the selected features and the current DNN hyperparameters.
   - Evaluate the DNN model's performance (eg. Classification task) on a validation dataset.
   - Assign fitness values to each wolf based on DNN performance (higher fitness for better results).
6. Identify the alpha, beta, and delta wolves (the top three with highest fitness).
7. Update the position of each wolf using the GWO equations:
   - Update wolves positions using equations 1,2,3,4.
   - For each wolf i:

i. Update position for α wolf using equation 1.

$$\alpha_\mathbf{new} = \alpha_\mathbf{old} + \mathbf{A} * (\mathbf{X} - \alpha_\mathbf{old}) \quad (1)$$

ii. Update position for β wolf using equation 2.

$$\beta_\mathbf{new} = \beta_\mathbf{old} + \mathbf{B} * (\mathbf{X} - \beta_\mathbf{old}) \quad (2)$$

iii. Update position for δ wolf using equation 3.

$$\boldsymbol{\delta_new} = \boldsymbol{\delta_old} + \boldsymbol{D} * (\boldsymbol{X} - \boldsymbol{\delta_old}) \quad (3)$$

iv. Update position for other-wolf (not alpha,beta or delta) us- ing equation 4.

$$\boldsymbol{Other_Wolf_new}\ \alpha_\boldsymbol{old} + \boldsymbol{A} * (\alpha_\boldsymbol{old} - \boldsymbol{Other_Wolf_old}) + \mathrm{Q}_\boldsymbol{old} + \boldsymbol{B} * (\mathrm{Q}\boldsymbol{old} - \boldsymbol{OtherWolfold}) + \boldsymbol{\delta old} + \boldsymbol{D} * (\boldsymbol{\delta old} - \boldsymbol{OtherWolfold}) \quad (4)$$

- Optimize DNN hyperparameters using positions of alpha, beta, and delta wolves:
- Decode positions of alpha, beta, and delta wolves to obtain hyperparameters
- Set DNN hyperparameters to the decoded values.

8. Update the current iteration count t=t+1.
9. DNN Training Phase: Train the DNN model using the selected features and optimized hyperparameters:

- Use the entire dataset (X, Y) for training.

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- Employ a gradient descent optimization method (Adam) to update the DNN's weights and biases.
- Monitor training progress and stop when convergence criteria are met or a maximum number of epochs is reached.

10. **Output:** Return the trained DNN model with the best-found hyperparameters and selected features.

Then, training of DNN is performed using the selected features. By this process, DNN takes reduced feature set as input and learns to classify medical images into relevant classes (e.g., disease vs. non-disease). For evaluation ChestX-ray14 dataset is considered. Evaluation of trained DNN model is performed using relevant metrics which is discussed in next section. Model performance is then assessed based on the selected features for classification task. Final step is to perform model evaluation using performance metrics as specified in Table I. Thus, GWO-DNN based approach enhances the process of image analysis by generating optimal features using GWO based on the image samples.

IV. Results And Discussions

The dataset used in proposed approach is ChestX-ray14 dataset from Kaggle. This dataset contains over 100,000 chest X-ray images across eight different pathology classes, including common thoracic diseases like pneumonia, tuberculosis, and lung cancer. Each image is labeled with multiple pathologies, allowing to perform multi-label classification tasks. The ChestX-ray14 dataset is commonly used in research related to chest radiology, deep learning, and medical image analysis. It's suitable to perform tasks such as disease classification, localization, and detection in chest X-ray images. Fig. 2 shows sample images from Chest X-ray dataset.

The proposed work integrates GWO with DNN for feature selection and classification in medical image analysis, GWO outputs an optimized feature subset, representing most relevant features for accurate classification.

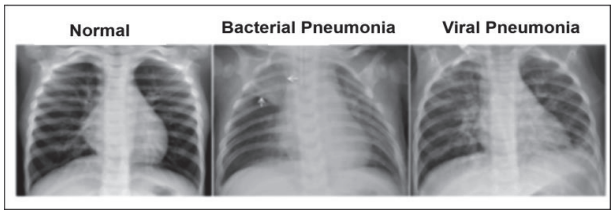


Fig. 2. Sample Images from ChestX-ray Dataset

This selected feature subset is then used as input for training and evaluating a DNN model. The final output of combined approach is an effective and optimized medical image classification model that utilizes feature selection power of GWO. This method enhances classification accuracy and helps in medical diagnosis and analysis. Performance evaluation of GWO-DNN model with relevant metrics is specified in Table I. The details reveal that the GWO-DNN model performs well compared to existing approaches in performing image analysis. The generated predictions by GWO-DNN model are represented in Fig. 3.

TABLE I. Comparative Analysis of Various Algorithms

Method	Dataset Used	Accuracy	Precision	Recall	F1-Score
GWO-DNN (Proposed)	ChestX-ray14	92.5	0.93	0.91	0.92
CNN[24]	ChestX-ray14	90.2	0.91	0.88	0.89
Multimodal[25]	LIDC-IDRI	88.7	0.89	0.87	0.88
Transfer Learning[26]	Skin Lesion	94.1	0.94	0.93	0.93
U-Net[27]	Brain MRI	91.3	0.92	0.90	0.91

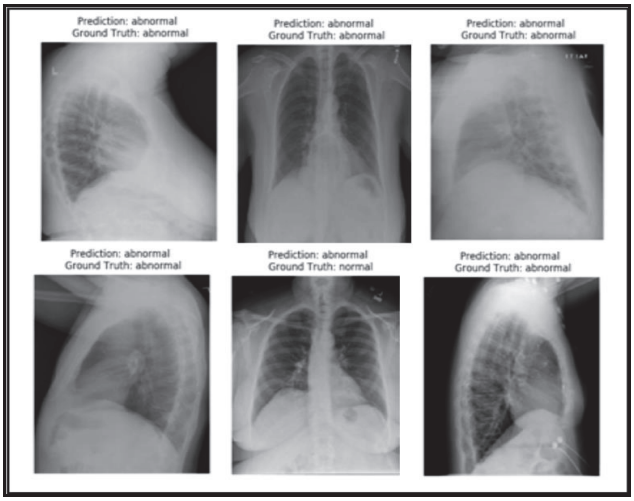


Fig. 3. Generated predictions by GWO-DNN model with Ground Truth Specification

V. Conclusion

Integration of Grey Wolf Optimizer (GWO) with Deep Neural Networks (DNNs) for feature selection and classification in medical image analysis has provided promising results and illustrate its capability as a transformative approach in healthcare sector. Experimental evaluation shown that GWO-DNN framework offers significant ad- vantages over conventional methods and standalone DNNs. It not only optimizes the selection of critical features within medical images but also enables creation of highly accurate and interpretable classification models. This integrated approach has the po- tential to significantly enhance diagnostic accuracy. This can be used as a valuable tool in various medical specialties, including radiology, pathology, and more. Future of the GWO-DNN framework lies in its potential to expand and adapt to various healthcare challenges. Further research could focus on enhancing interpretability, accommodating multi-modal data, promoting clinical validation and Scalability for larger datasets con- tributing to precise and efficient medical image analysis in various healthcare settings.

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